

Betting against beta

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Abstract

“Betting against beta” is a strategy used in investment. Investors are expected to generate positive abnormal return by shorting high beta stocks and going long low beta stocks. In this study, we are examining the funding constraint based explanation, the proposition 3, of betting against beta. We found that during periods of high funding constraint, the betting against beta effect becomes stronger.

Key Words:

Betting against beta, funding constraint

1. Introduction

Betting against beta is a finance strategy that Frazzini and Pedersen [\[1\]](#) use to generate abnormal returns. Investors take a short position in low beta stocks and at the same time take a long position in high beta stocks and they are expected to produce positive risk adjusted returns. This is the phenomenon known as the betting against beta anomaly. There are 5 propositions or predictions explained by the “Betting against beta” study: (1) Some investors, such as pension funds and mutual funds, facing difficulties in taking leverage would increase their allocation to high beta stock, hoping for higher return, when they form their portfolio. This increased demand for high beta stock would cause these stocks to become overvalued and subsequently underperform after the correction of this overvaluation. (2) A Betting against beta (BAB) factor produces significant positive risk adjusted returns. (3) When funding constraints tighten, the return of betting against the beta factor is low. (4) Increased funding liquidity risk would compress beta to 1. (5) constrained investors would hold risky assets.

In this study, we will explore the funding constraint based explanation of BAB factor. Funding constraints have become tightened recently after several interest rate hikes by the federal government. Therefore, according to the betting against beta model, we are expecting the betting against beta phenomenon to be stronger during periods of high funding constraint. We test 2 different types of funding constraints, the TED spread and the fed fund rate, and run regression based on basic CAPM and Fama French 3 factor models in our study. According to “Betting against beta”, we are expecting stocks with low beta will generate higher alpha than stocks with high beta.

2. Data and Methodology

2.1 Data

In order to replicate betting against beta factor, there are several crucial data we need to gather before our analysis. From the Ken French online library, we generated our monthly market data which includes market factor, size-related factor (*smb*), book-to-market related factor (*hml*), and monthly risk free rate. Our stock data is downloaded from the Center for Research in Security Prices (CRSP) database¹. We used the entire database to generate historical monthly stock information from 1972 to 2022, including monthly returns, prices, and number of shares outstanding. The exchange code, which indicates on which stock exchange a particular security is listed, is 1 for NYSE, 2 for Amex, and 3 for Nasdaq. To proceed with our project, We calculated the beta for each stock, based on this formula:

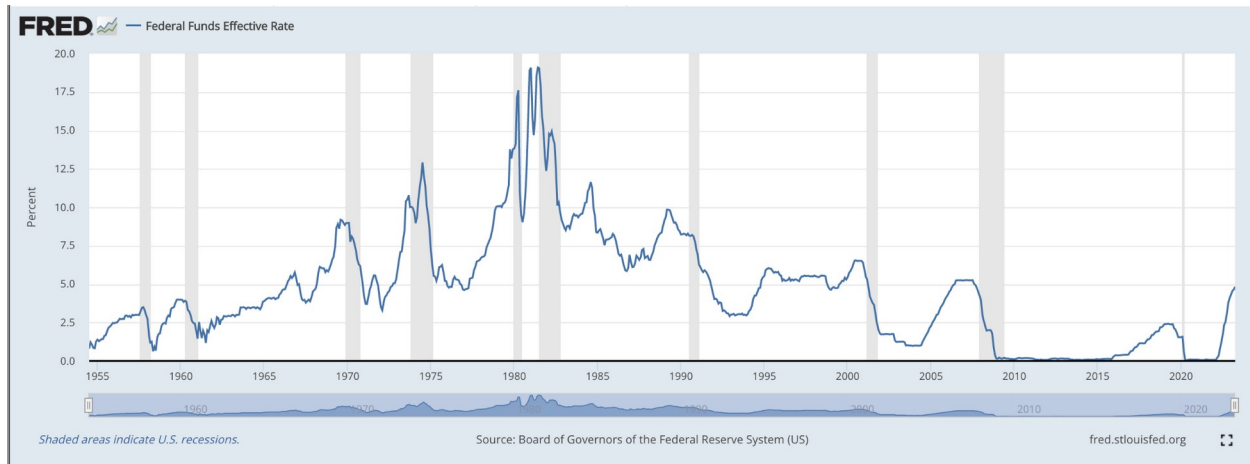
$$\beta_i = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$$

Moreover, our regression analysis includes 2 funding constraint factors for the funding-constraint-based explanation for the BAB effect, the TED Spread, in **figure 1**, (denoted as *month_tedrate*) and the fed fund rate, in **figure 2**, (denoted as *ffr*). We incorporated these two factors in the Fama French Three Factor which is expressed as:

$$R_{it} - R_{ft} = \alpha_{it} + \beta_1(R_{Mt} - R_{ft}) + \beta_2SMB_t + \beta_3HML_t$$

¹ *Note: the sources that we gathered our stock, market, and funding constraint data mentioned above are:
https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
<https://wrds-www.wharton.upenn.edu/>
<https://fred.stlouisfed.org/series/FEDFUNDS>
<https://fred.stlouisfed.org/series/TEDRATE>

Figure 1: Fed Fund Effective rate



*Note: the graph above shows the relationship between the year and its corresponding fed fund rate. The x-axis represent the year number, and Y-axis represents the fed fund rate in percentages

Figure 2: TED Spread



*Note: the graph above shows the relationship between the year and its corresponding TED Spread. The x-axis represent the year number, and Y-axis represents the TED Spread in percentage

The following **Table 1** shows definition and the name of each variable that we included in our regression analysis:

Table 1. Variables

date	The last trading date of the month
permno	5-digits permanent security ID
exchcd	exchange code 1 for NYSE, 2 for Amex, and 3 for Nasdaq
siccd	4-digits standard industry code
prc	Monthly price of each individual security
ret	Monthly return of each individual security
shrout	Numbers of shares outstanding of each individual security
retrf	Excess return of each individual security
me	Market capitalization of each stock
lme	1 month lagged market capitalization of each stock
beta	a stock's beta
lbeta	1 month lagged beta, used to generate portfolio
mktrf	Market excess return
sml	size related factor
hml	book-to-market related factor
rf	monthly risk free rate
ffr	monthly fed fund rate
lffr	1 month lagged fed fund rate, used to run regression analysis
M_tedrate	monthly ted spread
lM_tedrate	1 month lagged ted spread, used to run regression analysis

*Note: the table above shows name of each variable in our study and its corresponding definition

2.2 Methodology

2.2.1 Grouping beta

We grouped the data for all dates into five portfolios according to their beta size. The purpose of this is to distinguish whether each date is low beta or high beta. A portfolio belonging to 1 is low beta and a portfolio belonging to 5 is high beta. For example, if the first date has a beta less than 1-quantile, then the first date belongs to portfolio 1, which is low beta portfolio. If its beta is bigger than 4-quantile and less than 5-quantile, then it's belongs to portfolio 5, which is high beta portfolio.

2.2.2 Calculating value-weighted return and equal-weighted return

Value-weighted return takes into account the weight of each stock in the portfolio. Stocks with larger market capitalization have higher weights in the portfolio, while stocks with smaller market capitalization have lower weights. To calculate the value-weighted return for each stock, we first calculate the weight of each stock in the overall portfolio. This can be determined by the ratio of each stock's market value to the market value of the entire portfolio. We multiply the weight of each stock by its own return to get the value-weighted return. We use the same method to obtain value-weighted beta and value-weighted market cap.

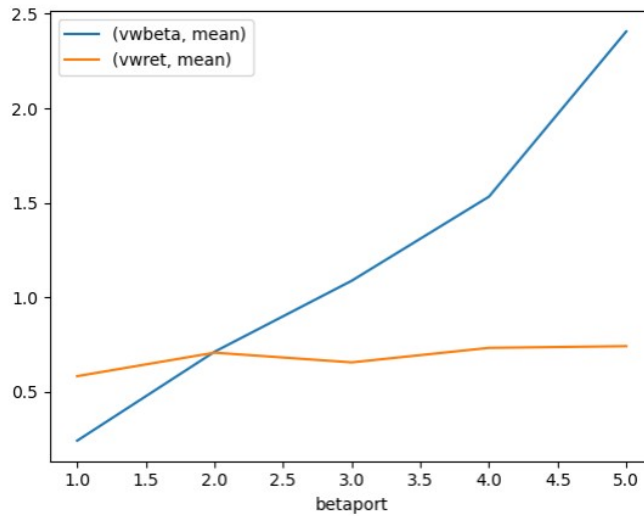
Moreover, we use the mean to calculate the equal-weighted return, equal-weighted beta, and equal-weighted market cap for each stock in the portfolio. Unlike value-weighted, equal-weighted return allocates the return of each stock in the portfolio equally, which each stock is equally weighted, regardless of the market capitalization or weight of the stock in the portfolio.

2.2.3 Building the High minus Low portfolio

We created a 6th portfolio, which is a high-beta portfolio (betaport 5) minus a low-beta portfolio (betaport 1). We performed group statistics and calculated each portfolio's mean, standard deviation, and t-value. However, our findings revealed that none of the t-statistical results for portfolio 6 were statistically significant for both value-weighted and equal-weighted returns. This observation suggests that the observed difference between the two portfolios may not be significant enough to eliminate the possibility of random error. Alternatively, this result could imply that the market may not accurately reflect the beta factor's impact on stock returns or that the sample data may not adequately capture the beta factor's true effect.

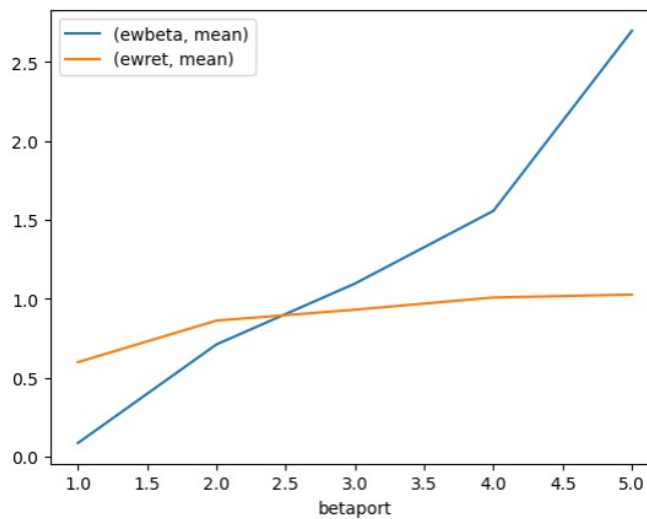
2.2.4 Add Market Data

Figure 3: Relationship between beta and value-weighted return



*Source:author's own photograph

Figure 4: Relationship between beta and equal-weighted return

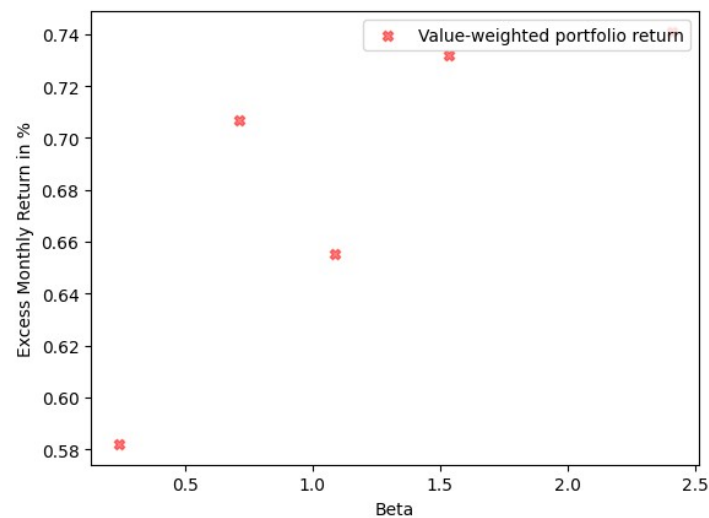


*Source:author's own photograph

Figure 3 illustrates the relationship between the beta and value-weighted return of the investment portfolio, and Figure 4 demonstrates the relationship between the beta and equal-weighted return. It is observed that as the value of beta increases, the average return of the investment portfolio shows a slightly increasing trend, indicating a positive correlation between the average return of

the portfolio and beta. However, this trend does not necessarily imply those high-beta portfolios are superior, as such portfolios often come with higher risks not adjusted for in the data points. Therefore, when making investment decisions, one needs to consider a combination of yield and risk, rather than just looking at the trend of one factor.

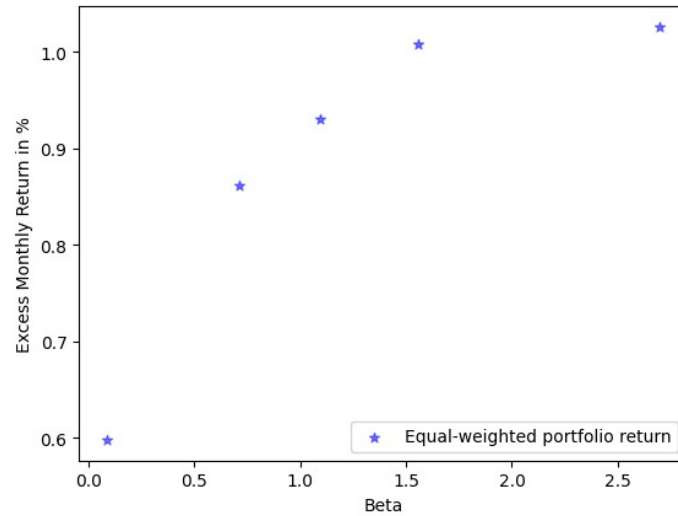
Figure 5: Relationship between beta and value-weighted excess monthly return



*Source:author's own photograph

Figure 5 shows the relationship between the beta and the value-weighted excess monthly return for each stock in the portfolio. By looking at the scatter plot distribution, we can see that the data points are relatively scattered, there is a large variation in the excess monthly return for stocks with different beta in the portfolio. Therefore, constructing a portfolio based on beta alone is not a completely accurate predictor of portfolio returns, and other factors need to be considered.

Figure 6: Relationship between beta and equal-weighted excess monthly return



*Source:author's own photograph

The horizontal coordinate of Figure 6 is the beta, and the vertical coordinate is the equal-weighted excess monthly return. Based on the distribution of scatter, it can be seen that equal-weighted return tends to increase slightly with increasing beta. However, the scatter distribution is relatively scattered, so the trend is not obvious. It is also important to note that these data are not risk-adjusted, i.e., they are not calibrated for risk. As the beta increases, the portfolio's risk also increases, which may weaken the positive correlation between beta and excess return.

Therefore, we add market data for the study, which are market excess return (mktrf), size-related factor (sml), and book-to-market related factor (hml). They are processed the same way as before so they can be merged into our existing data.

2.2.5 Set constant to 1

In the CAPM model, the expected return of the stock can be estimated by the following equation:

$$E(R_i) = R_f + \beta_i \times (E(R_m) - R_f)$$

Where $E(R_i)$ is the expected return of stock i , R_f is the risk-free rate, β_i is the market risk factor of stock i , and $E(R_m)$ is the expected market return rate. In the CAPM model, the constant term (α) is considered to be the risk-free interest rate R_f . Therefore, we set the constant term to 1, which is equivalent to expressing the expected excess return of a stock ($E(R_i) - R_f$) as a linear combination of β_i and the expected excess return of the market, where β_i represents the exposure of the stock relative to the market.

3. Results

3.1 value-weighted portfolio

In the first regression, we set x to value-weighted return and y to constant and market excess return. According to the regression results shown in Table 2, we find that the coefficient is negative and highly significant. Since we are using portfolio 6, the high beta portfolio minus the low beta portfolio, a negative coefficient means that the return for high beta is smaller than the return for low beta. This is consistent with the main point of betting against beta: investors can get higher risk-adjusted returns by going long on low-beta assets and short on high-beta assets.

Table 2. OLS results

	Constant (alpha)	mktrf
vwret	-0.0526	1.0408
	0.009	0.000
ewret	-0.25520	1.0324
	0.210	0.000

*Note: The first column of each OLS regression results is the coefficient and the second column is the P-value of t-test.

3.2 equal-weighted portfolio

In the second regression shown in Table 2, x is the equal-weighted return, and y is the constant and market excess return, which turns out to be insignificant. It may be because x is the equal-weighted return, and each stock is equally weighted, so the equal-weighted portfolio may have a smaller market capitalization and contain more stocks of smaller companies, and companies with larger market capitalization are a smaller proportion of the equal-weighted portfolio. This means that the equal-weighted portfolio may be more susceptible to fluctuations in the prices of the stocks of smaller companies. And if companies with more extensive market capitalization stock

price changes are relatively small, these changes may not be fully reflected in the equal-weighted portfolio, thus affecting the regression results.

3.3 With market factors

In addition, because only the constant term and the market excess return are included in the second regression, the model may not fully capture the impact of other factors in the equally weighted portfolio on returns. For example, other factors may include firm size, valuation, and style. Therefore, we will consider additional factors to continue analyzing the relationship between portfolio returns and beta.

Table 3. With mkt factor OLS results

	Constant (alpha)	mktrf	smb	hml
vwret+mkt factor	-0.4712	0.8726	0.7533	-0.3121
	0.006	0.000	0.000	0.000
ewret+mkt factor	-0.2919	0.8767	0.9014	-0.0634
	0.082	0.000	0.000	0.255

*Note: The first column of each OLS regression results is the coefficient and the second column is the P-value of t-test.

The new regression adds two market factors, size-related factor (sml) and book-to-market related factor (hml). From the table 3, we can see that the results of value-weighted and equal-weighted are negatively and significantly correlated. The regressions are always in portfolio 6, which means that the return of high beta is smaller than the return of low beta. It proves proposition 2 of betting against beta, that is, Betting against beta (BAB) factor, which is long leveraged low-beta assets and short high-beta assets, produces significant positive risk-adjusted returns.

3.4 monthly TED Spread

Using TED Spread to indicate the

Table 4. Ted spred OLS results

	Constant (alpha)	mktrf	smb	hml	lmonthly tedrate
vwret: tedrate	-1.0484	1.0768			1.0286
	0.009	0.000			0.074
ewret: tedrate	-0.9013	1.0979			1.3066
	0.025	0.000			0.025
vwret: tedrate + mkt factor	-0.8114	0.9249	0.7601	-0.2709	0.8245
	0.018	0.000	0.000	0.000	0.095
ewret: tedrate + mkt factor	-0.8187	0.9442	0.9424	-0.0277	1.3097
	0.014	0.000	0.000	0.675	0.007

*Note: The first column of each OLS regression results is the coefficient and the second column is the P-value of t-test.

3.5 monthly Fed Fund rate

When we turn into monthly fed fund rate (ffr), we can see in table 5 the results are not significant.

Table 5. ffr OLS results

	Constant (alpha)	mktrf	smb	hml	lffr
vwret: ffr	-0.2138	1.0380			-0.0669
	0.483	0.000			0.172
ewret: ffr	0.2046	1.0283			-0.0977
	0.501	0.000			0.046
vwret: ffr + mkt	-0.2115	0.8706	0.75538	-0.3086	-0.0558

factor	0.412	0.000	0.000	0.000	0.178
ewret: ffr + mkt factor	0.1501	0.8734	0.9024	-0.0574	-0.0950
	0.550	0.000	0.000	0.302	0.019

*Note: The first column of each OLS regression results is the coefficient and the second column is the P-value of t-test.

4. Conclusion

The results from our regression analysis indicate that the TED spread has a significant impact on the betting against beta factor. As the funding constraint factor, the TED spread, becomes stronger, investors tend to allocate more of their position in high beta stocks and generate lower abnormal returns. However, the other funding constraint factor, the fed fund rate, does not show a significant relationship with the BAB factor. This is probably due to the fact that people are not sensitive to this factor.

Based on the historical data that we gathered to generate our analysis we may conclude that the betting against beta anomaly can be explained by the funding constraint explanation. As cost of borrowing increases, we expect that investors will increase their position in high beta stocks and this increased demand will cause these stocks to be overvalued. In the future, this overvaluation will be subsequently corrected and therefore, the betting against beta factor will become stronger during periods of high funding constraint.

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